

|          |            |                                                                                                                 |    |     |
|----------|------------|-----------------------------------------------------------------------------------------------------------------|----|-----|
| <b>6</b> | <b>(a)</b> | $R = \rho l / A$                                                                                                | C1 |     |
|          |            | $A = [\pi \times (0.38 \times 10^{-3})^2] / 4 \quad (= 0.113 \times 10^{-6} \text{ m}^2)$                       | C1 |     |
|          |            | $R = (4.5 \times 10^{-7} \times 1.00) / ([\pi \times (0.38 \times 10^{-3})^2] / 4) = 4.0 \text{ (3.97)} \Omega$ | M1 | [3] |
|          | <b>(b)</b> | <b>(i)</b>                                                                                                      |    |     |
|          |            | $I = V / R$                                                                                                     | C1 |     |
|          |            | $= 2.0 / 5.0 = 0.4(0) \text{ A}$                                                                                | A1 | [2] |
|          |            | <b>(ii)</b>                                                                                                     |    |     |
|          |            | p.d. across BD $= 4 \times 0.4 = 1.6 \text{ V}$                                                                 | A1 | [1] |
|          |            | <b>(iii)</b>                                                                                                    |    |     |
|          |            | p.d. across BC ( $l$ ) $= 1.5 \text{ (V)}$                                                                      | C1 |     |
|          |            | $BC \text{ (} l \text{)} = (1.5 / 1.6) \times 100 = 94 \text{ (93.75) cm}$                                      | A1 | [2] |

| <b>Page 4</b> | <b>Mark Scheme</b>                                                | <b>Syllabus</b> | <b>Paper</b> |
|---------------|-------------------------------------------------------------------|-----------------|--------------|
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- (c) p.d. across wire not balancing e.m.f. of cell OR cell Y has current  
energy lost or lost volts due to internal resistance
- B1  
B1 [2]